IET, C-DAC

500 MCQ PRACTICE QUESTION BANK

DSA Using Java

Moderate to Difficult | Conceptual | Code Output | Error Analysis | Tricky Applications

Purpose

A subject-wise practice resource for engineering graduates preparing for C-DAC course examinations. Questions are freshly authored for practice and are not represented as actual or leaked C-DAC examination questions.

Coverage

Complexity analysis; arrays and linked lists; stacks and queues; recursion; trees and BST; heaps; searching and sorting; hashing; graph representations and traversals; shortest paths; spanning trees; algorithm design.

Instructions

Each question has four options and one best answer. Attempt the questions before consulting the consolidated answer key at the end. For code questions, assume standard language behavior and the stated code fragment is complete for the operation being tested.

Question count: 500 **Options:** A–D **Answer key:** Included

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QUESTION PAPER

DSA Using Java — 500 Questions

Q1.

A balanced binary search tree has 36 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 72
B. 6
C. 5
D. 35

Q2.

How many times does the inner statement execute?
for(int i=1;i<=22;i++) for(int j=1;j<=i;j++) work();

A. 462
B. 484
C. 253
D. 44

Q3.

A loop compares every pair of distinct elements exactly once in an array of 5 elements. How many comparisons occur?

A. 10
B. 9
C. 25
D. 15

Q4.

A balanced binary search tree has 12 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 11
C. 24
D. 3

Q5.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 8
B. 31
C. 16
D. 16

Q6.

An array-based circular queue has capacity 12. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 12
B. 11
C. 13
D. 24

Q7.

How many times does the inner statement execute?
for(int i=1;i<=23;i++) for(int j=1;j<=i;j++) work();

A. 276
B. 46
C. 506
D. 529

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Q8.

A balanced binary search tree has 27 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 26
B. 54
C. 4
D. 5

Q9.

What is the time complexity of accessing an array element by index?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)

Q10.

How many times does the inner statement execute?
for(int i=1;i<=24;i++) for(int j=1;j<=i;j++) work();

A. 300
B. 576
C. 552
D. 48

Q11.

A graph has 35 vertices and exactly 35 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 71
B. 70
C. 1225
D. 35

Q12.

Merge sort processes 24 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(1)
C. O(n^2)
D. O(log n)

Q13.

What is the worst-case lookup time in a hash table with all keys colliding into one chain?

A. O(n log n)
B. O(log n)
C. O(n)
D. O(1)

Q14.

A graph has 23 vertices and exactly 23 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 529
B. 47
C. 46
D. 23

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Q15.

An array-based circular queue has capacity 14. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 28
B. 15
C. 14
D. 13

Q16.

How many successful link updates are needed to reverse a singly linked list containing 22 nodes using the iterative three-pointer method?

A. 22
B. 23
C. 44
D. 21

Q17.

How many successful link updates are needed to reverse a singly linked list containing 25 nodes using the iterative three-pointer method?

A. 25
B. 50
C. 24
D. 26

Q18.

How many times does the inner statement execute?
for(int i=1;i<=9;i++) for(int j=1;j<=i;j++) work();

A. 81
B. 45
C. 18
D. 72

Q19.

How many times does the inner statement execute?
for(int i=1;i<=16;i++) for(int j=1;j<=i;j++) work();

A. 32
B. 136
C. 256
D. 240

Q20.

A loop compares every pair of distinct elements exactly once in an array of 12 elements. How many comparisons occur?

A. 65
B. 78
C. 66
D. 144

Q21.

How many times does the inner statement execute?
for(int i=1;i<=29;i++) for(int j=1;j<=i;j++) work();

A. 435
B. 812
C. 58
D. 841

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Q22.

How many successful link updates are needed to reverse a singly linked list containing 21 nodes using the iterative three-pointer method?

A. 42
B. 20
C. 21
D. 22

Q23.

Which traversal visits a node before its left and right subtrees?

A. Postorder
B. Inorder
C. Preorder
D. Reverse level order

Q24.

A balanced binary search tree has 35 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 34
B. 5
C. 6
D. 70

Q25.

How many times does the inner statement execute?
for(int i=1;i<=32;i++) for(int j=1;j<=i;j++) work();

A. 64
B. 992
C. 528
D. 1024

Q26.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 8
B. 16
C. 31
D. 16

Q27.

Merge sort processes 18 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n^2)
D. O(n log n)

Q28.

Merge sort processes 21 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n log n)
C. O(n^2)
D. O(log n)

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Q29.

A balanced binary search tree has 27 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 26
B. 54
C. 5
D. 4

Q30.

An array-based circular queue has capacity 33. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 34
B. 32
C. 33
D. 66

Q31.

A balanced binary search tree has 16 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 32
C. 5
D. 15

Q32.

Merge sort processes 6 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(log n)
D. O(1)

Q33.

A balanced binary search tree has 10 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 9
B. 20
C. 3
D. 4

Q34.

A graph has 20 vertices and exactly 20 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 400
B. 40
C. 20
D. 41

Q35.

Merge sort processes 16 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(1)
D. O(log n)

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Q36.

How many times does the inner statement execute?
for(int i=1;i<=33;i++) for(int j=1;j<=i;j++) work();

A. 66
B. 1056
C. 1089
D. 561

Q37.

How many times does the inner statement execute?
for(int i=1;i<=16;i++) for(int j=1;j<=i;j++) work();

A. 32
B. 256
C. 240
D. 136

Q38.

A balanced binary search tree has 35 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 70
B. 5
C. 34
D. 6

Q39.

An array-based circular queue has capacity 37. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 74
B. 37
C. 38
D. 36

Q40.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 7
C. 2
D. 4

Q41.

An array-based circular queue has capacity 21. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 21
B. 20
C. 22
D. 42

Q42.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 7
C. 2
D. 4

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Q43.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 15
B. 8
C. 9
D. 4

Q44.

Merge sort processes 16 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(log n)
C. O(1)
D. O(n^2)

Q45.

Merge sort processes 17 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(log n)
D. O(1)

Q46.

How many times does the inner statement execute?
for(int i=1;i<=26;i++) for(int j=1;j<=i;j++) work();

A. 52
B. 676
C. 650
D. 351

Q47.

How many successful link updates are needed to reverse a singly linked list containing 10 nodes using the iterative three-pointer method?

A. 11
B. 10
C. 20
D. 9

Q48.

What is the recurrence for binary search in the usual comparison model?

A. T(n)=2T(n/2)+O(1)
B. T(n)=T(n/2)+O(1)
C. T(n)=T(n^2)+O(1)
D. T(n)=T(n-1)+O(n)

Q49.

How many times does the inner statement execute?
for(int i=1;i<=20;i++) for(int j=1;j<=i;j++) work();

A. 400
B. 40
C. 210
D. 380

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Q50.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 32
B. 36
C. 127
D. 64

Q51.

An array-based circular queue has capacity 19. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 20
B. 38
C. 19
D. 18

Q52.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 15
B. 9
C. 8
D. 4

Q53.

How many times does the inner statement execute?
for(int i=1;i<=34;i++) for(int j=1;j<=i;j++) work();

A. 595
B. 68
C. 1156
D. 1122

Q54.

How many successful link updates are needed to reverse a singly linked list containing 5 nodes using the iterative three-pointer method?

A. 10
B. 6
C. 5
D. 4

Q55.

How many times does the inner statement execute?
for(int i=1;i<=28;i++) for(int j=1;j<=i;j++) work();

A. 784
B. 406
C. 56
D. 756

Q56.

An array-based circular queue has capacity 24. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 48
B. 23
C. 24
D. 25

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Q57.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 16
B. 32
C. 63
D. 25

Q58.

An array-based circular queue has capacity 40. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 40
B. 41
C. 39
D. 80

Q59.

A graph has 7 vertices and exactly 7 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 49
B. 7
C. 14
D. 15

Q60.

A loop compares every pair of distinct elements exactly once in an array of 34 elements. How many comparisons occur?

A. 595
B. 561
C. 1156
D. 560

Q61.

How many successful link updates are needed to reverse a singly linked list containing 34 nodes using the iterative three-pointer method?

A. 33
B. 35
C. 68
D. 34

Q62.

A loop compares every pair of distinct elements exactly once in an array of 14 elements. How many comparisons occur?

A. 90
B. 105
C. 196
D. 91

Q63.

A graph has 9 vertices and exactly 9 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 9
B. 19
C. 81
D. 18

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Q64.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 25
B. 63
C. 32
D. 16

Q65.

A balanced binary search tree has 26 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 25
B. 52
C. 5
D. 4

Q66.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 8
B. 4
C. 15
D. 9

Q67.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 4
B. 9
C. 15
D. 8

Q68.

How many times does the inner statement execute?
for(int i=1;i<=19;i++) for(int j=1;j<=i;j++) work();

A. 361
B. 38
C. 190
D. 342

Q69.

How many times does the inner statement execute?
for(int i=1;i<=24;i++) for(int j=1;j<=i;j++) work();

A. 576
B. 300
C. 48
D. 552

Q70.

A loop compares every pair of distinct elements exactly once in an array of 39 elements. How many comparisons occur?

A. 740
B. 741
C. 1521
D. 780

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Q71.

How many times does the inner statement execute?
for(int i=1;i<=34;i++) for(int j=1;j<=i;j++) work();

A. 1122
B. 595
C. 1156
D. 68

Q72.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 8
B. 4
C. 9
D. 15

Q73.

Merge sort processes 29 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(1)
C. O(log n)
D. O(n^2)

Q74.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 63
B. 32
C. 16
D. 25

Q75.

An array-based circular queue has capacity 24. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 48
B. 25
C. 24
D. 23

Q76.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 24
B. 11
C. 13
D. 12

Q77.

A loop compares every pair of distinct elements exactly once in an array of 13 elements. How many comparisons occur?

A. 91
B. 77
C. 169
D. 78

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Q78.

A balanced binary search tree has 38 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 37
B. 76
C. 5
D. 6

Q79.

A balanced binary search tree has 29 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 28
C. 4
D. 58

Q80.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 8
B. 9
C. 4
D. 15

Q81.

A loop compares every pair of distinct elements exactly once in an array of 23 elements. How many comparisons occur?

A. 529
B. 252
C. 253
D. 276

Q82.

How many successful link updates are needed to reverse a singly linked list containing 24 nodes using the iterative three-pointer method?

A. 23
B. 25
C. 24
D. 48

Q83.

An array-based circular queue has capacity 8. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 16
B. 9
C. 7
D. 8

Q84.

An array-based circular queue has capacity 10. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 9
B. 11
C. 10
D. 20

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Q85.

Merge sort processes 28 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(log n)
C. O(1)
D. O(n^2)

Q86.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 64
B. 127
C. 32
D. 36

Q87.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 24
B. 12
C. 13
D. 11

Q88.

How many successful link updates are needed to reverse a singly linked list containing 20 nodes using the iterative three-pointer method?

A. 40
B. 19
C. 21
D. 20

Q89.

A loop compares every pair of distinct elements exactly once in an array of 16 elements. How many comparisons occur?

A. 256
B. 136
C. 120
D. 119

Q90.

An array-based circular queue has capacity 40. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 41
B. 40
C. 39
D. 80

Q91.

How many successful link updates are needed to reverse a singly linked list containing 13 nodes using the iterative three-pointer method?

A. 12
B. 26
C. 13
D. 14

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Q92.

A balanced binary search tree has 24 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 48
C. 23
D. 4

Q93.

What does a stable sorting algorithm preserve?

A. The pivot position
B. Original array capacity
C. Relative order of equal-key elements
D. The number of comparisons

Q94.

A graph has 20 vertices and exactly 20 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 20
B. 40
C. 400
D. 41

Q95.

A loop compares every pair of distinct elements exactly once in an array of 25 elements. How many comparisons occur?

A. 325
B. 625
C. 299
D. 300

Q96.

A loop compares every pair of distinct elements exactly once in an array of 17 elements. How many comparisons occur?

A. 135
B. 136
C. 289
D. 153

Q97.

A graph has 14 vertices and exactly 14 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 29
B. 28
C. 14
D. 196

Q98.

A graph has 35 vertices and exactly 35 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 71
B. 35
C. 70
D. 1225

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Q99.

An array-based circular queue has capacity 32. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 31
B. 64
C. 32
D. 33

Q100.

How many successful link updates are needed to reverse a singly linked list containing 10 nodes using the iterative three-pointer method?

A. 9
B. 10
C. 20
D. 11

Q101.

How many successful link updates are needed to reverse a singly linked list containing 14 nodes using the iterative three-pointer method?

A. 13
B. 14
C. 28
D. 15

Q102.

A graph has 18 vertices and exactly 18 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 36
B. 324
C. 18
D. 37

Q103.

A graph has 25 vertices and exactly 25 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 51
B. 50
C. 25
D. 625

Q104.

Merge sort processes 5 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(1)
D. O(log n)

Q105.

A graph has 13 vertices and exactly 13 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 169
B. 26
C. 27
D. 13

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Q106.

A loop compares every pair of distinct elements exactly once in an array of 5 elements. How many comparisons occur?

A. 10
B. 15
C. 25
D. 9

Q107.

An array-based circular queue has capacity 25. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 50
B. 25
C. 24
D. 26

Q108.

An array-based circular queue has capacity 22. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 23
B. 21
C. 44
D. 22

Q109.

How many times does the inner statement execute?
for(int i=1;i<=15;i++) for(int j=1;j<=i;j++) work();

A. 120
B. 30
C. 225
D. 210

Q110.

An array-based circular queue has capacity 22. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 21
B. 23
C. 22
D. 44

Q111.

A loop compares every pair of distinct elements exactly once in an array of 31 elements. How many comparisons occur?

A. 465
B. 496
C. 961
D. 464

Q112.

How many times does the inner statement execute?
for(int i=1;i<=15;i++) for(int j=1;j<=i;j++) work();

A. 210
B. 30
C. 120
D. 225

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Q113.

How many successful link updates are needed to reverse a singly linked list containing 37 nodes using the iterative three-pointer method?

A. 37
B. 74
C. 36
D. 38

Q114.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 4
C. 7
D. 2

Q115.

An array-based circular queue has capacity 19. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 38
B. 19
C. 20
D. 18

Q116.

A loop compares every pair of distinct elements exactly once in an array of 19 elements. How many comparisons occur?

A. 361
B. 170
C. 171
D. 190

Q117.

A graph has 11 vertices and exactly 11 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 11
B. 23
C. 22
D. 121

Q118.

How many times does the inner statement execute?
for(int i=1;i<=28;i++) for(int j=1;j<=i;j++) work();

A. 784
B. 756
C. 56
D. 406

Q119.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 32
B. 127
C. 36
D. 64

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Q120.

A loop compares every pair of distinct elements exactly once in an array of 38 elements. How many comparisons occur?

A. 1444
B. 702
C. 703
D. 741

Q121.

A balanced binary search tree has 37 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 74
B. 6
C. 36
D. 5

Q122.

A balanced binary search tree has 38 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 6
C. 37
D. 76

Q123.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 13
B. 12
C. 24
D. 11

Q124.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 2
B. 4
C. 7
D. 4

Q125.

A graph has 37 vertices and exactly 37 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 37
B. 74
C. 1369
D. 75

Q126.

Merge sort processes 24 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(n log n)
D. O(1)

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Q127.

A graph has 15 vertices and exactly 15 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 15
B. 30
C. 31
D. 225

Q128.

A balanced binary search tree has 33 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 32
B. 5
C. 6
D. 66

Q129.

A balanced binary search tree has 22 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 44
B. 21
C. 4
D. 5

Q130.

A balanced binary search tree has 20 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 19
B. 5
C. 40
D. 4

Q131.

How many successful link updates are needed to reverse a singly linked list containing 15 nodes using the iterative three-pointer method?

A. 15
B. 16
C. 14
D. 30

Q132.

A balanced binary search tree has 6 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 12
B. 2
C. 3
D. 5

Q133.

Merge sort processes 18 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(log n)
C. O(n^2)
D. O(1)

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Q134.

How many successful link updates are needed to reverse a singly linked list containing 36 nodes using the iterative three-pointer method?

A. 35
B. 37
C. 72
D. 36

Q135.

An array-based circular queue has capacity 24. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 48
B. 25
C. 23
D. 24

Q136.

Merge sort processes 31 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(log n)
D. O(1)

Q137.

A loop compares every pair of distinct elements exactly once in an array of 13 elements. How many comparisons occur?

A. 169
B. 78
C. 77
D. 91

Q138.

A loop compares every pair of distinct elements exactly once in an array of 34 elements. How many comparisons occur?

A. 560
B. 595
C. 1156
D. 561

Q139.

A loop compares every pair of distinct elements exactly once in an array of 36 elements. How many comparisons occur?

A. 629
B. 666
C. 1296
D. 630

Q140.

A graph has 5 vertices and exactly 5 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 11
B. 25
C. 5
D. 10

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Q141.

How many times does the inner statement execute?
for(int i=1;i<=30;i++) for(int j=1;j<=i;j++) work();

A. 60
B. 870
C. 465
D. 900

Q142.

Merge sort processes 27 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(1)
C. O(n^2)
D. O(log n)

Q143.

Merge sort processes 13 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(log n)
D. O(1)

Q144.

An array-based circular queue has capacity 6. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 7
B. 12
C. 5
D. 6

Q145.

Merge sort processes 31 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(1)
C. O(n log n)
D. O(log n)

Q146.

A loop compares every pair of distinct elements exactly once in an array of 20 elements. How many comparisons occur?

A. 190
B. 189
C. 400
D. 210

Q147.

A graph has 15 vertices and exactly 15 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 30
B. 15
C. 31
D. 225

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Q148.

How many times does the inner statement execute?
for(int i=1;i<=16;i++) for(int j=1;j<=i;j++) work();

A. 240
B. 136
C. 32
D. 256

Q149.

How many successful link updates are needed to reverse a singly linked list containing 11 nodes using the iterative three-pointer method?

A. 22
B. 11
C. 12
D. 10

Q150.

A graph has 30 vertices and exactly 30 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 60
B. 900
C. 61
D. 30

Q151.

What does topological ordering require?

A. A directed acyclic graph
B. A weighted tree only
C. A graph with exactly one cycle
D. An undirected complete graph

Q152.

How many successful link updates are needed to reverse a singly linked list containing 8 nodes using the iterative three-pointer method?

A. 8
B. 7
C. 16
D. 9

Q153.

How many successful link updates are needed to reverse a singly linked list containing 31 nodes using the iterative three-pointer method?

A. 31
B. 30
C. 32
D. 62

Q154.

Merge sort processes 27 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(1)
C. O(n log n)
D. O(log n)

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Q155.

A graph has 14 vertices and exactly 14 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 29
B. 196
C. 14
D. 28

Q156.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 127
B. 64
C. 32
D. 36

Q157.

Which collision resolution technique stores colliding entries in a bucket structure at each table index?

A. Double hashing
B. Quadratic probing
C. Linear probing
D. Separate chaining

Q158.

A graph has 8 vertices and exactly 8 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 16
B. 64
C. 8
D. 17

Q159.

An array-based circular queue has capacity 19. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 38
B. 18
C. 19
D. 20

Q160.

How many times does the inner statement execute?
for(int i=1;i<=27;i++) for(int j=1;j<=i;j++) work();

A. 54
B. 378
C. 729
D. 702

Q161.

How many successful link updates are needed to reverse a singly linked list containing 25 nodes using the iterative three-pointer method?

A. 50
B. 26
C. 24
D. 25

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Q162.

A loop compares every pair of distinct elements exactly once in an array of 27 elements. How many comparisons occur?

A. 729
B. 378
C. 351
D. 350

Q163.

Merge sort processes 5 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(1)
D. O(n log n)

Q164.

Merge sort processes 32 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(1)
C. O(n log n)
D. O(log n)

Q165.

How many successful link updates are needed to reverse a singly linked list containing 13 nodes using the iterative three-pointer method?

A. 13
B. 26
C. 14
D. 12

Q166.

A loop compares every pair of distinct elements exactly once in an array of 17 elements. How many comparisons occur?

A. 135
B. 153
C. 136
D. 289

Q167.

A graph has 6 vertices and exactly 6 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 12
B. 6
C. 36
D. 13

Q168.

A loop compares every pair of distinct elements exactly once in an array of 32 elements. How many comparisons occur?

A. 1024
B. 496
C. 495
D. 528

Q169.

Merge sort processes 22 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(1)
C. O(log n)
D. O(n log n)

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Q170.

A graph has 40 vertices and exactly 40 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 81
B. 80
C. 40
D. 1600

Q171.

A loop compares every pair of distinct elements exactly once in an array of 21 elements. How many comparisons occur?

A. 209
B. 441
C. 231
D. 210

Q172.

How many successful link updates are needed to reverse a singly linked list containing 37 nodes using the iterative three-pointer method?

A. 74
B. 38
C. 37
D. 36

Q173.

How many times does the inner statement execute?
for(int i=1;i<=31;i++) for(int j=1;j<=i;j++) work();

A. 930
B. 961
C. 496
D. 62

Q174.

A balanced binary search tree has 39 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 38
C. 78
D. 6

Q175.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 2
C. 7
D. 4

Q176.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 16
B. 31
C. 8
D. 16

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Q177.

Merge sort processes 30 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n log n)
D. O(n^2)

Q178.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 31
B. 8
C. 16
D. 16

Q179.

A balanced binary search tree has 32 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 31
B. 64
C. 5
D. 6

Q180.

How many successful link updates are needed to reverse a singly linked list containing 38 nodes using the iterative three-pointer method?

A. 76
B. 39
C. 38
D. 37

Q181.

A loop compares every pair of distinct elements exactly once in an array of 30 elements. How many comparisons occur?

A. 434
B. 435
C. 465
D. 900

Q182.

Merge sort processes 5 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(log n)
C. O(n log n)
D. O(n^2)

Q183.

How many successful link updates are needed to reverse a singly linked list containing 25 nodes using the iterative three-pointer method?

A. 50
B. 25
C. 26
D. 24

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Q184.

Merge sort processes 5 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n log n)
C. O(n^2)
D. O(1)

Q185.

Merge sort processes 29 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n log n)
D. O(n^2)

Q186.

An array-based circular queue has capacity 27. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 54
B. 28
C. 27
D. 26

Q187.

A loop compares every pair of distinct elements exactly once in an array of 24 elements. How many comparisons occur?

A. 276
B. 576
C. 300
D. 275

Q188.

A graph has 9 vertices and exactly 9 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 81
B. 18
C. 19
D. 9

Q189.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 4
C. 2
D. 7

Q190.

An array-based circular queue has capacity 8. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 16
B. 7
C. 9
D. 8

Q191.

A balanced binary search tree has 18 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 36
C. 5
D. 17

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Q192.

How many successful link updates are needed to reverse a singly linked list containing 22 nodes using the iterative three-pointer method?

A. 44
B. 23
C. 22
D. 21

Q193.

A balanced binary search tree has 33 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 6
C. 66
D. 32

Q194.

Which traversal of a binary search tree visits keys in nondecreasing order?

A. Preorder
B. Inorder
C. Postorder
D. Level order

Q195.

How many times does the inner statement execute?
for(int i=1;i<=8;i++) for(int j=1;j<=i;j++) work();

A. 36
B. 64
C. 56
D. 16

Q196.

How many times does the inner statement execute?
for(int i=1;i<=22;i++) for(int j=1;j<=i;j++) work();

A. 253
B. 462
C. 44
D. 484

Q197.

An array-based circular queue has capacity 40. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 80
B. 40
C. 39
D. 41

Q198.

An array-based circular queue has capacity 22. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 21
B. 44
C. 22
D. 23

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Q199.

How many successful link updates are needed to reverse a singly linked list containing 29 nodes using the iterative three-pointer method?

A. 30
B. 29
C. 58
D. 28

Q200.

Merge sort processes 28 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(log n)
D. O(1)

Q201.

An array-based circular queue has capacity 7. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 6
B. 7
C. 8
D. 14

Q202.

Merge sort processes 14 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n^2)
C. O(1)
D. O(n log n)

Q203.

How many successful link updates are needed to reverse a singly linked list containing 36 nodes using the iterative three-pointer method?

A. 36
B. 72
C. 35
D. 37

Q204.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 4
B. 8
C. 9
D. 15

Q205.

How many successful link updates are needed to reverse a singly linked list containing 25 nodes using the iterative three-pointer method?

A. 25
B. 26
C. 24
D. 50

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Q206.

A balanced binary search tree has 31 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 30
B. 4
C. 62
D. 5

Q207.

How many times does the inner statement execute?
for(int i=1;i<=29;i++) for(int j=1;j<=i;j++) work();

A. 812
B. 58
C. 841
D. 435

Q208.

Merge sort processes 16 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(1)
D. O(log n)

Q209.

A balanced binary search tree has 25 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 24
C. 4
D. 50

Q210.

How many successful link updates are needed to reverse a singly linked list containing 9 nodes using the iterative three-pointer method?

A. 8
B. 9
C. 18
D. 10

Q211.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 25
B. 32
C. 16
D. 63

Q212.

Merge sort processes 27 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n log n)
C. O(n^2)
D. O(1)

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Q213.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 31
B. 16
C. 16
D. 8

Q214.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 32
B. 16
C. 25
D. 63

Q215.

A graph has 13 vertices and exactly 13 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 27
B. 26
C. 13
D. 169

Q216.

A balanced binary search tree has 23 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 46
C. 4
D. 22

Q217.

A balanced binary search tree has 7 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 3
B. 14
C. 2
D. 6

Q218.

A graph has 5 vertices and exactly 5 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 10
B. 25
C. 11
D. 5

Q219.

What is the maximum number of nodes at level k of a binary tree when the root is level 0?

A. k+1
B. 2^k
C. 2k
D. k^2

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Q220.

How many successful link updates are needed to reverse a singly linked list containing 13 nodes using the iterative three-pointer method?

A. 13
B. 26
C. 12
D. 14

Q221.

A graph has 23 vertices and exactly 23 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 47
B. 46
C. 23
D. 529

Q222.

Which data structure supports efficient prefix lookup over strings?

A. Stack
B. Trie
C. Disjoint-set forest
D. Binary heap

Q223.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 2
B. 4
C. 7
D. 4

Q224.

How many times does the inner statement execute?
for(int i=1;i<=6;i++) for(int j=1;j<=i;j++) work();

A. 12
B. 36
C. 30
D. 21

Q225.

A graph has 9 vertices and exactly 9 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 18
B. 81
C. 9
D. 19

Q226.

A graph has 28 vertices and exactly 28 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 784
B. 56
C. 57
D. 28

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Q227.

An array-based circular queue has capacity 34. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 68
B. 33
C. 35
D. 34

Q228.

A balanced binary search tree has 34 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 6
C. 68
D. 33

Q229.

Which algorithm finds shortest paths from a source in a graph with non-negative edge weights?

A. Kruskal's algorithm
B. DFS
C. Prim's algorithm
D. Dijkstra's algorithm

Q230.

Merge sort processes 26 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(1)
D. O(log n)

Q231.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 9
B. 4
C. 15
D. 8

Q232.

A balanced binary search tree has 20 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 19
B. 40
C. 5
D. 4

Q233.

Which heap property holds in a min-heap?

A. Every left child is smaller than every right child
B. The array is fully sorted
C. Each parent key is no greater than its children
D. The root is always the maximum

Q234.

A loop compares every pair of distinct elements exactly once in an array of 7 elements. How many comparisons occur?

A. 20
B. 49
C. 28
D. 21

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Q235.

How many times does the inner statement execute?
for(int i=1;i<=23;i++) for(int j=1;j<=i;j++) work();

A. 506
B. 46
C. 529
D. 276

Q236.

A loop compares every pair of distinct elements exactly once in an array of 22 elements. How many comparisons occur?

A. 230
B. 484
C. 253
D. 231

Q237.

A graph has 34 vertices and exactly 34 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 68
B. 1156
C. 69
D. 34

Q238.

A graph has 37 vertices and exactly 37 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 1369
B. 37
C. 75
D. 74

Q239.

A loop compares every pair of distinct elements exactly once in an array of 37 elements. How many comparisons occur?

A. 666
B. 665
C. 1369
D. 703

Q240.

How many times does the inner statement execute?
for(int i=1;i<=15;i++) for(int j=1;j<=i;j++) work();

A. 120
B. 225
C. 210
D. 30

Q241.

A loop compares every pair of distinct elements exactly once in an array of 19 elements. How many comparisons occur?

A. 171
B. 170
C. 361
D. 190

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Q242.

Merge sort processes 22 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n log n)
C. O(log n)
D. O(n^2)

Q243.

A loop compares every pair of distinct elements exactly once in an array of 6 elements. How many comparisons occur?

A. 21
B. 15
C. 14
D. 36

Q244.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 32
B. 127
C. 64
D. 36

Q245.

How many times does the inner statement execute?
for(int i=1;i<=13;i++) for(int j=1;j<=i;j++) work();

A. 26
B. 91
C. 169
D. 156

Q246.

Merge sort processes 33 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n^2)
C. O(n log n)
D. O(1)

Q247.

How many successful link updates are needed to reverse a singly linked list containing 15 nodes using the iterative three-pointer method?

A. 15
B. 14
C. 30
D. 16

Q248.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 11
B. 13
C. 24
D. 12

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Q249.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 2
B. 4
C. 7
D. 4

Q250.

A balanced binary search tree has 39 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 38
B. 78
C. 5
D. 6

Q251.

A balanced binary search tree has 15 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 3
C. 14
D. 30

Q252.

How many times does the inner statement execute?
for(int i=1;i<=40;i++) for(int j=1;j<=i;j++) work();

A. 1600
B. 80
C. 820
D. 1560

Q253.

An array-based circular queue has capacity 25. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 26
B. 50
C. 25
D. 24

Q254.

How many successful link updates are needed to reverse a singly linked list containing 17 nodes using the iterative three-pointer method?

A. 34
B. 17
C. 18
D. 16

Q255.

An array-based circular queue has capacity 26. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 52
B. 25
C. 27
D. 26

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Q256.

How many times does the inner statement execute?
for(int i=1;i<=19;i++) for(int j=1;j<=i;j++) work();

A. 38
B. 190
C. 361
D. 342

Q257.

A graph has 8 vertices and exactly 8 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 16
B. 64
C. 17
D. 8

Q258.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 4
B. 15
C. 9
D. 8

Q259.

An array-based circular queue has capacity 14. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 15
B. 28
C. 13
D. 14

Q260.

A loop compares every pair of distinct elements exactly once in an array of 23 elements. How many comparisons occur?

A. 252
B. 253
C. 276
D. 529

Q261.

An array-based circular queue has capacity 39. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 39
B. 78
C. 40
D. 38

Q262.

A balanced binary search tree has 29 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 58
B. 5
C. 28
D. 4

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Q263.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 9
B. 15
C. 8
D. 4

Q264.

How many times does the inner statement execute?
for(int i=1;i<=23;i++) for(int j=1;j<=i;j++) work();

A. 529
B. 276
C. 506
D. 46

Q265.

A loop compares every pair of distinct elements exactly once in an array of 20 elements. How many comparisons occur?

A. 400
B. 190
C. 210
D. 189

Q266.

How many successful link updates are needed to reverse a singly linked list containing 26 nodes using the iterative three-pointer method?

A. 27
B. 52
C. 26
D. 25

Q267.

How many times does the inner statement execute?
for(int i=1;i<=24;i++) for(int j=1;j<=i;j++) work();

A. 48
B. 300
C. 576
D. 552

Q268.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 32
B. 25
C. 16
D. 63

Q269.

What is the height of a single-node tree when height is counted in edges?

A. 2
B. -1
C. 0
D. 1

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Q270.

Merge sort processes 7 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(1)
C. O(log n)
D. O(n^2)

Q271.

How many successful link updates are needed to reverse a singly linked list containing 33 nodes using the iterative three-pointer method?

A. 66
B. 33
C. 32
D. 34

Q272.

How many times does the inner statement execute?
for(int i=1;i<=21;i++) for(int j=1;j<=i;j++) work();

A. 441
B. 231
C. 420
D. 42

Q273.

An array-based circular queue has capacity 7. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 14
B. 7
C. 8
D. 6

Q274.

A loop compares every pair of distinct elements exactly once in an array of 7 elements. How many comparisons occur?

A. 49
B. 21
C. 20
D. 28

Q275.

An array-based circular queue has capacity 15. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 14
B. 30
C. 15
D. 16

Q276.

An array-based circular queue has capacity 29. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 28
B. 29
C. 30
D. 58

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Q277.

A graph has 35 vertices and exactly 35 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 70
B. 35
C. 1225
D. 71

Q278.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 16
B. 31
C. 16
D. 8

Q279.

Merge sort processes 11 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(log n)
D. O(1)

Q280.

An array-based circular queue has capacity 25. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 26
B. 24
C. 25
D. 50

Q281.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n^2)
C. O(1)
D. O(n log n)

Q282.

A balanced binary search tree has 27 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 26
B. 5
C. 54
D. 4

Q283.

Merge sort processes 14 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n^2)
C. O(n log n)
D. O(1)

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Q284.

A balanced binary search tree has 38 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 76
B. 5
C. 37
D. 6

Q285.

Which structure supports efficient disjoint-set union and find operations?

A. Binary search tree only
B. Deque
C. Disjoint-set union (union-find)
D. Circular queue

Q286.

How many successful link updates are needed to reverse a singly linked list containing 10 nodes using the iterative three-pointer method?

A. 20
B. 10
C. 9
D. 11

Q287.

An array-based circular queue has capacity 25. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 26
B. 50
C. 25
D. 24

Q288.

Merge sort processes 17 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(log n)
D. O(1)

Q289.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 63
B. 25
C. 32
D. 16

Q290.

A loop compares every pair of distinct elements exactly once in an array of 17 elements. How many comparisons occur?

A. 136
B. 135
C. 153
D. 289

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Q291.

A balanced binary search tree has 10 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 3
B. 20
C. 4
D. 9

Q292.

A graph has 34 vertices and exactly 34 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 69
B. 1156
C. 68
D. 34

Q293.

How many times does the inner statement execute?
for(int i=1;i<=5;i++) for(int j=1;j<=i;j++) work();

A. 15
B. 25
C. 20
D. 10

Q294.

Merge sort processes 40 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n log n)
D. O(n^2)

Q295.

A graph has 26 vertices and exactly 26 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 676
B. 53
C. 52
D. 26

Q296.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n log n)
C. O(n^2)
D. O(1)

Q297.

Merge sort processes 11 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n log n)
C. O(log n)
D. O(n^2)

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Q298.

A loop compares every pair of distinct elements exactly once in an array of 31 elements. How many comparisons occur?

A. 464
B. 496
C. 961
D. 465

Q299.

How many successful link updates are needed to reverse a singly linked list containing 6 nodes using the iterative three-pointer method?

A. 12
B. 6
C. 5
D. 7

Q300.

An array-based circular queue has capacity 29. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 30
B. 58
C. 29
D. 28

Q301.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 2
C. 4
D. 7

Q302.

How many successful link updates are needed to reverse a singly linked list containing 9 nodes using the iterative three-pointer method?

A. 18
B. 8
C. 10
D. 9

Q303.

How many times does the inner statement execute?
for(int i=1;i<=31;i++) for(int j=1;j<=i;j++) work();

A. 496
B. 961
C. 62
D. 930

Q304.

How many successful link updates are needed to reverse a singly linked list containing 9 nodes using the iterative three-pointer method?

A. 9
B. 8
C. 10
D. 18

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Q305.

How many times does the inner statement execute?
for(int i=1;i<=7;i++) for(int j=1;j<=i;j++) work();

A. 14
B. 42
C. 28
D. 49

Q306.

A loop compares every pair of distinct elements exactly once in an array of 31 elements. How many comparisons occur?

A. 464
B. 465
C. 961
D. 496

Q307.

Merge sort processes 6 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n log n)
C. O(n^2)
D. O(log n)

Q308.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(n log n)
C. O(1)
D. O(log n)

Q309.

A graph has 36 vertices and exactly 36 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 36
B. 1296
C. 72
D. 73

Q310.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 8
B. 16
C. 31
D. 16

Q311.

How many successful link updates are needed to reverse a singly linked list containing 28 nodes using the iterative three-pointer method?

A. 29
B. 56
C. 27
D. 28

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Q312.

A graph has 37 vertices and exactly 37 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 75
B. 1369
C. 37
D. 74

Q313.

How many times does the inner statement execute?
for(int i=1;i<=18;i++) for(int j=1;j<=i;j++) work();

A. 36
B. 171
C. 306
D. 324

Q314.

A loop compares every pair of distinct elements exactly once in an array of 10 elements. How many comparisons occur?

A. 44
B. 45
C. 100
D. 55

Q315.

An array-based circular queue has capacity 8. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 7
B. 16
C. 9
D. 8

Q316.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 16
B. 32
C. 63
D. 25

Q317.

A graph has 6 vertices and exactly 6 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 6
B. 13
C. 36
D. 12

Q318.

Merge sort processes 18 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(log n)
C. O(1)
D. O(n^2)

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Q319.

Merge sort processes 9 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n^2)
D. O(n log n)

Q320.

How many successful link updates are needed to reverse a singly linked list containing 33 nodes using the iterative three-pointer method?

A. 32
B. 66
C. 33
D. 34

Q321.

A balanced binary search tree has 30 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 4
C. 60
D. 29

Q322.

A loop compares every pair of distinct elements exactly once in an array of 29 elements. How many comparisons occur?

A. 405
B. 841
C. 435
D. 406

Q323.

How many times does the inner statement execute?
for(int i=1;i<=35;i++) for(int j=1;j<=i;j++) work();

A. 1225
B. 630
C. 1190
D. 70

Q324.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 127
B. 32
C. 36
D. 64

Q325.

A loop compares every pair of distinct elements exactly once in an array of 34 elements. How many comparisons occur?

A. 561
B. 595
C. 560
D. 1156

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Q326.

A balanced binary search tree has 34 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 68
C. 33
D. 6

Q327.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 8
B. 15
C. 9
D. 4

Q328.

A loop compares every pair of distinct elements exactly once in an array of 22 elements. How many comparisons occur?

A. 231
B. 484
C. 253
D. 230

Q329.

What does DFS commonly use to remember the next vertex to explore?

A. A circular array queue only
B. A stack or recursion call stack
C. A min-heap
D. A hash map only

Q330.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(log n)
D. O(1)

Q331.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 9
B. 15
C. 4
D. 8

Q332.

How many times does the inner statement execute?
for(int i=1;i<=26;i++) for(int j=1;j<=i;j++) work();

A. 676
B. 351
C. 52
D. 650

Q333.

Which graph traversal uses a queue in its standard implementation?

A. Dijkstra's relaxation only
B. DFS
C. BFS
D. Topological DFS only

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Q334.

A balanced binary search tree has 40 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 80
B. 5
C. 6
D. 39

Q335.

What is the space complexity of an adjacency matrix for V vertices?

A. O(V)
B. O(E log V)
C. O(V^2)
D. O(V+E)

Q336.

How many times does the inner statement execute?
for(int i=1;i<=31;i++) for(int j=1;j<=i;j++) work();

A. 961
B. 62
C. 496
D. 930

Q337.

How many successful link updates are needed to reverse a singly linked list containing 33 nodes using the iterative three-pointer method?

A. 32
B. 66
C. 34
D. 33

Q338.

Which technique is used by union-find to improve operation efficiency?

A. Path compression and union by rank/size
B. Memoized recursion only
C. Heapify
D. Pivot partitioning

Q339.

Merge sort processes 20 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(n log n)
D. O(1)

Q340.

How many times does the inner statement execute?
for(int i=1;i<=18;i++) for(int j=1;j<=i;j++) work();

A. 171
B. 324
C. 306
D. 36

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Q341.

A loop compares every pair of distinct elements exactly once in an array of 36 elements. How many comparisons occur?

A. 666
B. 630
C. 1296
D. 629

Q342.

An array-based circular queue has capacity 17. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 18
B. 17
C. 16
D. 34

Q343.

How many successful link updates are needed to reverse a singly linked list containing 15 nodes using the iterative three-pointer method?

A. 16
B. 15
C. 30
D. 14

Q344.

A balanced binary search tree has 8 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 16
B. 7
C. 3
D. 4

Q345.

A loop compares every pair of distinct elements exactly once in an array of 5 elements. How many comparisons occur?

A. 15
B. 25
C. 9
D. 10

Q346.

A loop compares every pair of distinct elements exactly once in an array of 32 elements. How many comparisons occur?

A. 528
B. 496
C. 495
D. 1024

Q347.

How many times does the inner statement execute?
for(int i=1;i<=15;i++) for(int j=1;j<=i;j++) work();

A. 210
B. 225
C. 30
D. 120

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Q348.

Which algorithmic paradigm does merge sort use?

A. Greedy only
B. Divide and conquer
C. Backtracking only
D. Branch and bound

Q349.

What is the worst-case time complexity of insertion sort?

A. O(log n)
B. O(n^2)
C. O(n log n)
D. O(1)

Q350.

How many successful link updates are needed to reverse a singly linked list containing 17 nodes using the iterative three-pointer method?

A. 34
B. 16
C. 18
D. 17

Q351.

Which algorithm is commonly used to find a minimum spanning tree and sorts edges by weight?

A. Bellman-Ford
B. Kruskal's algorithm
C. Floyd-Warshall
D. Binary search

Q352.

A balanced binary search tree has 13 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 12
B. 4
C. 26
D. 3

Q353.

Which algorithm can detect a negative-weight cycle reachable from a source?

A. Binary search
B. Bellman-Ford
C. Dijkstra
D. Prim

Q354.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 8
B. 31
C. 16
D. 16

Q355.

Merge sort processes 5 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n^2)
C. O(log n)
D. O(n log n)

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Q356.

A graph has 18 vertices and exactly 18 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 36
B. 18
C. 324
D. 37

Q357.

An array-based circular queue has capacity 28. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 29
B. 27
C. 28
D. 56

Q358.

How many successful link updates are needed to reverse a singly linked list containing 31 nodes using the iterative three-pointer method?

A. 30
B. 31
C. 62
D. 32

Q359.

How many times does the inner statement execute?
for(int i=1;i<=34;i++) for(int j=1;j<=i;j++) work();

A. 68
B. 1122
C. 1156
D. 595

Q360.

An array-based circular queue has capacity 15. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 15
B. 30
C. 14
D. 16

Q361.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 7
B. 4
C. 2
D. 4

Q362.

An array-based circular queue has capacity 19. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 38
B. 18
C. 19
D. 20

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Q363.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 63
B. 32
C. 25
D. 16

Q364.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 9
B. 4
C. 15
D. 8

Q365.

How many times does the inner statement execute?
for(int i=1;i<=28;i++) for(int j=1;j<=i;j++) work();

A. 56
B. 784
C. 756
D. 406

Q366.

A graph has 39 vertices and exactly 39 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 1521
B. 78
C. 79
D. 39

Q367.

A balanced binary search tree has 40 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 80
B. 6
C. 5
D. 39

Q368.

A graph has 10 vertices and exactly 10 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 100
B. 10
C. 21
D. 20

Q369.

An array-based circular queue has capacity 32. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 32
B. 33
C. 31
D. 64

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Q370.

In a full binary tree, if every level from 0 through 2 is completely filled, how many leaves are at the deepest level?

A. 4
B. 4
C. 2
D. 7

Q371.

A graph has 40 vertices and exactly 40 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 1600
B. 80
C. 40
D. 81

Q372.

A loop compares every pair of distinct elements exactly once in an array of 10 elements. How many comparisons occur?

A. 100
B. 55
C. 45
D. 44

Q373.

An array-based circular queue has capacity 31. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 62
B. 32
C. 31
D. 30

Q374.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 16
B. 31
C. 16
D. 8

Q375.

A loop compares every pair of distinct elements exactly once in an array of 22 elements. How many comparisons occur?

A. 230
B. 231
C. 253
D. 484

Q376.

How many times does the inner statement execute?
for(int i=1;i<=33;i++) for(int j=1;j<=i;j++) work();

A. 1056
B. 1089
C. 66
D. 561

Q377.

What is the average-case search complexity of a well-distributed hash table?

A. O(n^2)
B. O(log n)
C. O(1)
D. O(n log n)

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Q378.

How many times does the inner statement execute?
for(int i=1;i<=5;i++) for(int j=1;j<=i;j++) work();

A. 25
B. 10
C. 15
D. 20

Q379.

A graph has 16 vertices and exactly 16 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 256
B. 32
C. 16
D. 33

Q380.

How many successful link updates are needed to reverse a singly linked list containing 16 nodes using the iterative three-pointer method?

A. 17
B. 16
C. 32
D. 15

Q381.

A balanced binary search tree has 29 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 58
C. 28
D. 4

Q382.

A graph has 19 vertices and exactly 19 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 39
B. 19
C. 361
D. 38

Q383.

A balanced binary search tree has 35 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 34
B. 70
C. 5
D. 6

Q384.

A balanced binary search tree has 15 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 30
C. 14
D. 3

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Q385.

How many times does the inner statement execute?
for(int i=1;i<=20;i++) for(int j=1;j<=i;j++) work();

A. 210
B. 40
C. 400
D. 380

Q386.

What is the amortized time complexity of append in a dynamic array?

A. O(1)
B. O(n) for every append
C. O(log n)
D. O(n log n)

Q387.

An array-based circular queue has capacity 34. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 68
B. 35
C. 34
D. 33

Q388.

An array-based circular queue has capacity 9. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 10
B. 18
C. 9
D. 8

Q389.

An array-based circular queue has capacity 6. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 12
B. 6
C. 7
D. 5

Q390.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 64
B. 127
C. 36
D. 32

Q391.

A loop compares every pair of distinct elements exactly once in an array of 18 elements. How many comparisons occur?

A. 152
B. 171
C. 324
D. 153

Q392.

An array-based circular queue has capacity 30. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 31
B. 60
C. 29
D. 30

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Q393.

How many successful link updates are needed to reverse a singly linked list containing 25 nodes using the iterative three-pointer method?

A. 25
B. 50
C. 24
D. 26

Q394.

Which structure is most suitable for checking balanced parentheses?

A. Min-heap
B. Stack
C. Adjacency matrix
D. Queue

Q395.

A graph has 29 vertices and exactly 29 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 58
B. 59
C. 841
D. 29

Q396.

A loop compares every pair of distinct elements exactly once in an array of 13 elements. How many comparisons occur?

A. 77
B. 78
C. 169
D. 91

Q397.

A loop compares every pair of distinct elements exactly once in an array of 27 elements. How many comparisons occur?

A. 729
B. 350
C. 378
D. 351

Q398.

An array-based circular queue has capacity 33. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 34
B. 33
C. 66
D. 32

Q399.

How many successful link updates are needed to reverse a singly linked list containing 35 nodes using the iterative three-pointer method?

A. 34
B. 70
C. 35
D. 36

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Q400.

An array-based circular queue has capacity 32. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 31
B. 33
C. 64
D. 32

Q401.

A graph has 33 vertices and exactly 33 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 1089
B. 66
C. 67
D. 33

Q402.

A loop compares every pair of distinct elements exactly once in an array of 19 elements. How many comparisons occur?

A. 190
B. 171
C. 361
D. 170

Q403.

How many successful link updates are needed to reverse a singly linked list containing 38 nodes using the iterative three-pointer method?

A. 37
B. 76
C. 39
D. 38

Q404.

A balanced binary search tree has 25 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 24
B. 4
C. 50
D. 5

Q405.

A loop compares every pair of distinct elements exactly once in an array of 32 elements. How many comparisons occur?

A. 496
B. 528
C. 495
D. 1024

Q406.

A balanced binary search tree has 17 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 16
B. 4
C. 5
D. 34

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Q407.

A loop compares every pair of distinct elements exactly once in an array of 9 elements. How many comparisons occur?

A. 81
B. 35
C. 36
D. 45

Q408.

A loop compares every pair of distinct elements exactly once in an array of 8 elements. How many comparisons occur?

A. 28
B. 36
C. 64
D. 27

Q409.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 11
B. 12
C. 24
D. 13

Q410.

An array-based circular queue has capacity 32. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 31
B. 33
C. 64
D. 32

Q411.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 32
B. 63
C. 16
D. 25

Q412.

A balanced binary search tree has 21 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 20
C. 5
D. 42

Q413.

An array-based circular queue has capacity 8. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 9
B. 7
C. 8
D. 16

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Q414.

How many times does the inner statement execute?
for(int i=1;i<=11;i++) for(int j=1;j<=i;j++) work();

A. 22
B. 110
C. 66
D. 121

Q415.

A graph has 13 vertices and exactly 13 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 13
B. 26
C. 27
D. 169

Q416.

A loop compares every pair of distinct elements exactly once in an array of 29 elements. How many comparisons occur?

A. 435
B. 405
C. 841
D. 406

Q417.

How many times does the inner statement execute?
for(int i=1;i<=6;i++) for(int j=1;j<=i;j++) work();

A. 12
B. 30
C. 36
D. 21

Q418.

Merge sort processes 14 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(1)
D. O(n log n)

Q419.

Merge sort processes 22 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n^2)
C. O(n log n)
D. O(log n)

Q420.

A graph has 18 vertices and exactly 18 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 37
B. 18
C. 36
D. 324

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Q421.

A balanced binary search tree has 38 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 37
B. 6
C. 76
D. 5

Q422.

How many successful link updates are needed to reverse a singly linked list containing 38 nodes using the iterative three-pointer method?

A. 38
B. 39
C. 76
D. 37

Q423.

A loop compares every pair of distinct elements exactly once in an array of 20 elements. How many comparisons occur?

A. 189
B. 400
C. 190
D. 210

Q424.

An array-based circular queue has capacity 17. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 17
B. 16
C. 18
D. 34

Q425.

Merge sort processes 16 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(1)
D. O(n log n)

Q426.

A balanced binary search tree has 14 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 13
B. 4
C. 28
D. 3

Q427.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 32
B. 25
C. 16
D. 63

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Q428.

A loop compares every pair of distinct elements exactly once in an array of 20 elements. How many comparisons occur?

A. 190
B. 400
C. 210
D. 189

Q429.

What is the worst-case time complexity of binary search on a sorted array?

A. O(n log n)
B. O(1)
C. O(log n)
D. O(n)

Q430.

A loop compares every pair of distinct elements exactly once in an array of 39 elements. How many comparisons occur?

A. 1521
B. 740
C. 741
D. 780

Q431.

How many times does the inner statement execute?
for(int i=1;i<=29;i++) for(int j=1;j<=i;j++) work();

A. 841
B. 58
C. 812
D. 435

Q432.

A graph has 6 vertices and exactly 6 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 36
B. 13
C. 12
D. 6

Q433.

How many successful link updates are needed to reverse a singly linked list containing 22 nodes using the iterative three-pointer method?

A. 21
B. 23
C. 44
D. 22

Q434.

Merge sort processes 38 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(n^2)
C. O(1)
D. O(log n)

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Q435.

A graph has 7 vertices and exactly 7 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 7
B. 15
C. 14
D. 49

Q436.

Which notation describes an asymptotic upper bound?

A. Big-Omega
B. Big-Theta only
C. Big-O
D. Little-omega

Q437.

A loop compares every pair of distinct elements exactly once in an array of 8 elements. How many comparisons occur?

A. 27
B. 64
C. 28
D. 36

Q438.

What is the worst-case time complexity of quicksort with consistently poor pivot choices?

A. O(n)
B. O(n log n)
C. O(n^2)
D. O(log n)

Q439.

A balanced binary search tree has 13 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 3
B. 26
C. 12
D. 4

Q440.

How many successful link updates are needed to reverse a singly linked list containing 24 nodes using the iterative three-pointer method?

A. 23
B. 24
C. 25
D. 48

Q441.

A graph has 34 vertices and exactly 34 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 34
B. 69
C. 68
D. 1156

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Q442.

A graph has 11 vertices and exactly 11 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 121
B. 11
C. 23
D. 22

Q443.

An array-based circular queue has capacity 9. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 9
B. 18
C. 10
D. 8

Q444.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(n log n)
D. O(1)

Q445.

An array-based circular queue has capacity 16. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 17
B. 16
C. 15
D. 32

Q446.

How many successful link updates are needed to reverse a singly linked list containing 40 nodes using the iterative three-pointer method?

A. 39
B. 41
C. 80
D. 40

Q447.

A graph has 38 vertices and exactly 38 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 77
B. 38
C. 76
D. 1444

Q448.

An array-based circular queue has capacity 36. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 35
B. 37
C. 36
D. 72

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Q449.

A balanced binary search tree has 5 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 3
B. 2
C. 4
D. 10

Q450.

A balanced binary search tree has 38 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 76
B. 6
C. 5
D. 37

Q451.

How many times does the inner statement execute?
for(int i=1;i<=32;i++) for(int j=1;j<=i;j++) work();

A. 528
B. 1024
C. 992
D. 64

Q452.

How many times does the inner statement execute?
for(int i=1;i<=5;i++) for(int j=1;j<=i;j++) work();

A. 20
B. 15
C. 10
D. 25

Q453.

A balanced binary search tree has 21 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 20
C. 4
D. 42

Q454.

A balanced binary search tree has 31 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 62
C. 5
D. 30

Q455.

A loop compares every pair of distinct elements exactly once in an array of 7 elements. How many comparisons occur?

A. 28
B. 21
C. 20
D. 49

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Q456.

Merge sort processes 11 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(1)
C. O(n^2)
D. O(n log n)

Q457.

A graph has 12 vertices and exactly 12 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 12
B. 144
C. 25
D. 24

Q458.

Merge sort processes 7 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n log n)
B. O(1)
C. O(log n)
D. O(n^2)

Q459.

How many times does the inner statement execute?
for(int i=1;i<=34;i++) for(int j=1;j<=i;j++) work();

A. 1156
B. 68
C. 595
D. 1122

Q460.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 31
B. 8
C. 16
D. 16

Q461.

In a full binary tree, if every level from 0 through 5 is completely filled, how many leaves are at the deepest level?

A. 25
B. 63
C. 32
D. 16

Q462.

A balanced binary search tree has 12 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 4
B. 11
C. 24
D. 3

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Q463.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 16
B. 31
C. 8
D. 16

Q464.

A balanced binary search tree has 15 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 30
B. 4
C. 3
D. 14

Q465.

How many successful link updates are needed to reverse a singly linked list containing 12 nodes using the iterative three-pointer method?

A. 24
B. 11
C. 12
D. 13

Q466.

Merge sort processes 8 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(1)
C. O(log n)
D. O(n log n)

Q467.

How many successful link updates are needed to reverse a singly linked list containing 17 nodes using the iterative three-pointer method?

A. 34
B. 17
C. 18
D. 16

Q468.

Which data structure follows LIFO order?

A. Deque only
B. Priority queue
C. Queue
D. Stack

Q469.

An array-based circular queue has capacity 28. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 28
B. 56
C. 27
D. 29

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Q470.

A graph has 26 vertices and exactly 26 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 53
B. 676
C. 26
D. 52

Q471.

Merge sort processes 34 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(log n)
B. O(n^2)
C. O(n log n)
D. O(1)

Q472.

An array-based circular queue has capacity 8. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 7
B. 9
C. 16
D. 8

Q473.

A balanced binary search tree has 35 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 34
B. 6
C. 70
D. 5

Q474.

A graph has 22 vertices and exactly 22 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 45
B. 22
C. 484
D. 44

Q475.

Which sorting algorithm is stable in its standard implementation and has O(n log n) worst-case time?

A. Selection sort
B. Merge sort
C. Heap sort
D. Classic in-place quicksort

Q476.

A graph has 12 vertices and exactly 12 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 12
B. 25
C. 24
D. 144

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Q477.

Which operation is O(1) in a singly linked list when a pointer to the head is available?

A. Insert after the kth node without traversal
B. Access the kth element
C. Search an arbitrary value
D. Insert at head

Q478.

How many successful link updates are needed to reverse a singly linked list containing 19 nodes using the iterative three-pointer method?

A. 20
B. 19
C. 38
D. 18

Q479.

In a full binary tree, if every level from 0 through 4 is completely filled, how many leaves are at the deepest level?

A. 16
B. 31
C. 8
D. 16

Q480.

Which sorting algorithm repeatedly selects the minimum remaining element and places it next?

A. Insertion sort
B. Counting sort
C. Merge sort
D. Selection sort

Q481.

A balanced binary search tree has 28 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 5
B. 4
C. 56
D. 27

Q482.

What is the worst-case time complexity of searching in an unbalanced BST?

A. O(log n)
B. O(1)
C. O(n)
D. O(log log n)

Q483.

Merge sort processes 33 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n log n)
C. O(n^2)
D. O(log n)

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Q484.

A graph has 24 vertices and exactly 24 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 24
B. 48
C. 576
D. 49

Q485.

An array-based circular queue has capacity 16. If one slot is reserved to distinguish full from empty, what is its usable capacity?

A. 16
B. 15
C. 17
D. 32

Q486.

Which data structure is commonly used to implement a priority queue efficiently?

A. Unsorted linked list only
B. Stack
C. Binary heap
D. Trie

Q487.

A balanced binary search tree has 15 nodes. What is the maximum number of edges on a root-to-leaf path if its height is bounded by floor(log2(n))?

A. 30
B. 3
C. 4
D. 14

Q488.

A graph has 12 vertices and exactly 12 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 144
B. 12
C. 25
D. 24

Q489.

How many times does the inner statement execute?
for(int i=1;i<=14;i++) for(int j=1;j<=i;j++) work();

A. 105
B. 196
C. 182
D. 28

Q490.

A loop compares every pair of distinct elements exactly once in an array of 24 elements. How many comparisons occur?

A. 576
B. 276
C. 300
D. 275

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Q491.

A graph has 29 vertices and exactly 29 edges. In an adjacency-list representation of an undirected graph, how many adjacency entries are stored?

A. 841
B. 59
C. 29
D. 58

Q492.

How many times does the inner statement execute?
for(int i=1;i<=5;i++) for(int j=1;j<=i;j++) work();

A. 20
B. 15
C. 25
D. 10

Q493.

A loop compares every pair of distinct elements exactly once in an array of 17 elements. How many comparisons occur?

A. 153
B. 289
C. 136
D. 135

Q494.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 4
B. 15
C. 9
D. 8

Q495.

In a full binary tree, if every level from 0 through 6 is completely filled, how many leaves are at the deepest level?

A. 64
B. 127
C. 32
D. 36

Q496.

In a full binary tree, if every level from 0 through 3 is completely filled, how many leaves are at the deepest level?

A. 4
B. 8
C. 9
D. 15

Q497.

Merge sort processes 13 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(n^2)
B. O(log n)
C. O(1)
D. O(n log n)

Q498.

Which linked-list type allows traversal in both directions?

A. Circular singly list only
B. Doubly linked list
C. Singly linked list
D. Skip list with one level

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Q499.

Merge sort processes 19 elements. Which expression best describes its asymptotic comparison-time bound?

A. O(1)
B. O(n^2)
C. O(log n)
D. O(n log n)

Q500.

How many times does the inner statement execute?
for(int i=1;i<=33;i++) for(int j=1;j<=i;j++) work();

A. 66
B. 561
C. 1056
D. 1089

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CONSOLIDATED ANSWER KEY

Answers are listed by question number. Each key corresponds to the option order shown in the question section.

1. C	2. C	3. A	4. D	5. D
6. B	7. A	8. C	9. A	10. A
11. B	12. A	13. C	14. C	15. D
16. D	17. C	18. B	19. B	20. C
21. A	22. B	23. C	24. B	25. C
26. B	27. D	28. B	29. D	30. B
31. A	32. A	33. C	34. B	35. A
36. D	37. D	38. B	39. D	40. A
41. B	42. A	43. B	44. A	45. B
46. D	47. D	48. B	49. C	50. D
51. D	52. C	53. A	54. D	55. B
56. B	57. B	58. C	59. C	60. B
61. A	62. D	63. D	64. C	65. D
66. A	67. D	68. C	69. B	70. B
71. B	72. A	73. A	74. B	75. D
76. B	77. D	78. C	79. C	80. A
81. C	82. A	83. C	84. A	85. A
86. A	87. D	88. B	89. C	90. C
91. A	92. D	93. C	94. B	95. D
96. B	97. B	98. C	99. A	100. A
101. A	102. A	103. B	104. A	105. B
106. A	107. C	108. B	109. A	110. A
111. A	112. C	113. C	114. B	115. D
116. C	117. C	118. D	119. D	120. C
121. D	122. A	123. D	124. D	125. B
126. C	127. B	128. B	129. C	130. D
131. C	132. B	133. A	134. A	135. C
136. A	137. B	138. D	139. D	140. D
141. C	142. A	143. B	144. C	145. C
146. A	147. A	148. B	149. D	150. A
151. A	152. B	153. B	154. C	155. D

156. B	157. D	158. A	159. B	160. B
161. C	162. C	163. D	164. C	165. D
166. C	167. A	168. B	169. D	170. B
171. D	172. D	173. C	174. A	175. A
176. A	177. C	178. D	179. C	180. D
181. B	182. C	183. D	184. B	185. C
186. D	187. A	188. B	189. B	190. B
191. A	192. D	193. A	194. B	195. A
196. A	197. C	198. A	199. D	200. B
201. A	202. D	203. C	204. B	205. C
206. B	207. D	208. B	209. C	210. A
211. B	212. B	213. B	214. A	215. B
216. C	217. C	218. A	219. B	220. C
221. B	222. B	223. B	224. D	225. A
226. B	227. B	228. A	229. D	230. B
231. D	232. D	233. C	234. D	235. D
236. D	237. A	238. D	239. A	240. A
241. A	242. B	243. B	244. C	245. B
246. C	247. B	248. A	249. B	250. C
251. B	252. C	253. D	254. D	255. B
256. B	257. A	258. D	259. C	260. B
261. D	262. D	263. C	264. B	265. B
266. D	267. B	268. A	269. C	270. A
271. C	272. B	273. D	274. B	275. A
276. A	277. A	278. C	279. A	280. B
281. D	282. D	283. C	284. B	285. C
286. C	287. D	288. A	289. C	290. A
291. A	292. C	293. A	294. C	295. C
296. B	297. B	298. D	299. C	300. D
301. C	302. B	303. A	304. B	305. C
306. B	307. B	308. B	309. C	310. B
311. C	312. D	313. B	314. B	315. A
316. B	317. D	318. A	319. D	320. A

321. B	322. D	323. B	324. D	325. A
326. A	327. A	328. A	329. B	330. A
331. D	332. B	333. C	334. B	335. C
336. C	337. A	338. A	339. C	340. A
341. B	342. C	343. D	344. C	345. D
346. B	347. D	348. B	349. B	350. B
351. B	352. D	353. B	354. D	355. D
356. A	357. B	358. A	359. D	360. C
361. B	362. B	363. B	364. D	365. D
366. B	367. C	368. D	369. C	370. B
371. B	372. C	373. D	374. A	375. B
376. D	377. C	378. C	379. B	380. D
381. D	382. D	383. C	384. D	385. A
386. A	387. D	388. D	389. D	390. A
391. D	392. C	393. C	394. B	395. A
396. B	397. D	398. D	399. A	400. A
401. B	402. B	403. A	404. B	405. A
406. B	407. C	408. A	409. A	410. A
411. A	412. A	413. B	414. C	415. B
416. D	417. D	418. D	419. C	420. C
421. D	422. D	423. C	424. B	425. D
426. D	427. A	428. A	429. C	430. C
431. D	432. C	433. A	434. A	435. C
436. C	437. C	438. C	439. A	440. A
441. C	442. D	443. D	444. C	445. C
446. A	447. C	448. A	449. B	450. C
451. A	452. B	453. C	454. A	455. B
456. D	457. D	458. A	459. C	460. C
461. C	462. D	463. A	464. C	465. B
466. D	467. D	468. D	469. C	470. D
471. C	472. A	473. D	474. D	475. B
476. C	477. D	478. D	479. D	480. D
481. B	482. C	483. B	484. B	485. B

486. C	**487. B**	**488. D**	**489. A**	**490. B**
491. D	**492. B**	**493. C**	**494. D**	**495. A**
496. B	**497. D**	**498. B**	**499. D**	**500. B**

Reference basis: C-DAC official PG-DAC/PGCP-AC course brochure and publicly available module outlines for topic coverage; publicly available C-DAC evaluation guidance for the objective-question format and difficulty framework. Questions in this booklet are independently authored practice items, not copied past-paper questions.

Official course outline: https://cdac.gov.in/index.aspx?courseid=0&id=DAC
C-DAC evaluation guideline reference: publicly available C-DAC evaluation guideline PDF (search result/source reference used during preparation).

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